



Fig. 2: Yellow connections are rat lines

puter. Most of the symbols come built-in with EDA tools which you just have to select, place on the worksheet and connect with each other. And if some symbol is missing, it can be easily created and used in all future designs. The complete design itself can be simulated and checked for expected functionality, which reduces the chance of errors at later stages. Such organisation and automation, in turn, reduces human error and increases the overall productivity.

Now, to make the PCB of your design, you will have to draw it manually, corresponding to the schematic, if EDA tools were not available—something today's designers cannot even imagine doing. It is highly unproductive and there are high chances of human error. With EDA tools, there is now a lot of automation involved for PCB design. You can import your design directly to the PCB tool and you will find all the footprints already connected with rat lines. These rat lines exactly correspond to the schematic's wiring. This means that if your schematic is correct, the PCB will match it exactly. A lot of automation features like auto-routing, auto-placement, DRC (design rule check), etc, help in making the highest-quality, error-free PCBs.

Once the PCB is ready, you will have to send it to the PCB manufac-

turer for prototyping. With these EDA tools, the manufacturer and designer are in sync. The standard output data to manufacture the PCB can be easily generated. This is then directly fed to the machines for manufacturing. After you receive the PCBs, they will be sent to the assembler for the mounting of components. The Bill of material (BOM) and manufacturing data required by the assembler is created with just the click of a button and the chip shooter mounts the components in minutes. If you have to do all this manually, the time to market will be long, and the results uncertain.

Why is it important to select the right EDA tool?

While selecting an EDA tool, the first thing that you should keep in mind is that this tool will be used throughout the organisation by different departments. Therefore this tool should support the requirements of each department. You would not want to use several different tools because then, inter-departmental communication becomes impossible. So it is best that you consider the requirements of each department and choose the tool that suits them all or can be configured to do so.

It requires considerable training when implementing a new tool in your organisation. Only after proper train-

ing and a lot of practice can the tool be used to its fullest. But training comes at a price and for practice, you will have to invest a lot of time. Also, if you buy a proprietary tool, the licence is quite expensive. This makes it really important that you select the right EDA tool the first time itself.

Open source or proprietary?

With open source tools, you can save yourself the expense of buying a licence and you do not stand to lose anything apart from time even if your selection is not correct the first time. And if you get this right, you stand to gain a lot. Open source software is made available to the public, enabling anyone to use it without paying royalties or fees. Open source software evolves through community cooperation with the sharing of experiences and ideas that can then be incorporated into future releases.

Some people still prefer proprietary software because they believe that open source software is never stable. To some extent, they are right, because open source software doesn't come with a guarantee that the software will continue to be developed with regular upgrades. Also, it takes a lot of time to evaluate whether such software meets all the requirements of an organisation; after which, if it is not appropriate, you have to start the evaluation process all over again with another software. In the case of proprietary software, the vendor takes the responsibility of continuity, features and training.

But with the open source community growing very fast and producing reliable products, the trust level is also rising. These communities have created thousands of stable products which are used by millions of happy users. Also, the support provided by the community is amazing because it is not from one company but enthusiasts all over the world that are always happy to help.

The developments in open source software happen faster than in the proprietary world because the ideas